Practice assignment 8 - Not Graded

This assignment will not be graded and is only for practice.
Note:Note: Suppose W_1 and W_2 W1 and W2 are subspaces of a vector space V V . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.$$ W1 + W2 = {w1 + w2 ∣ w1 ∈ W1, w2 ∈ W2}.

One can verify that $W_1 + W_2$ W1 + W2 is a subspace of V V .

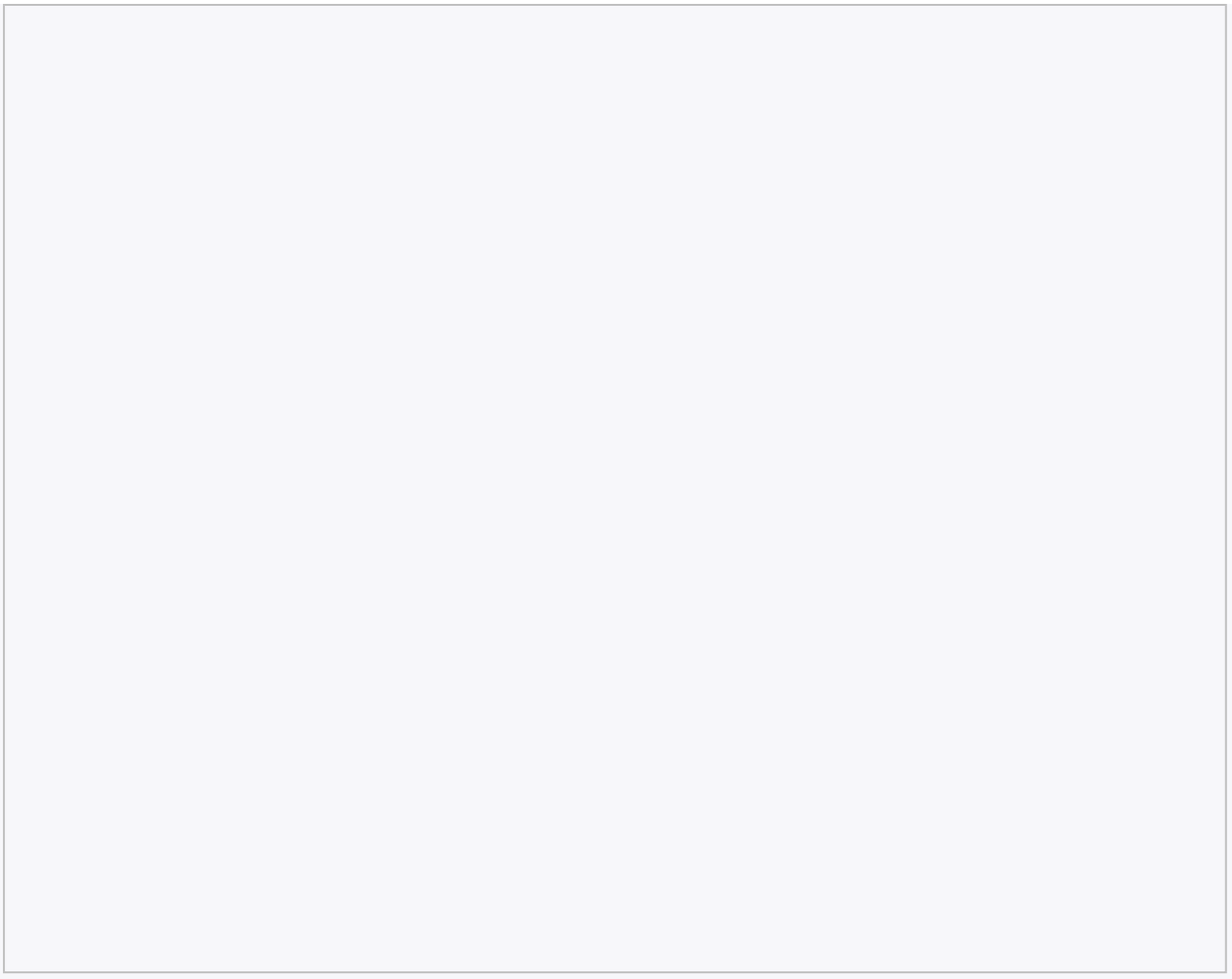
Multiple Choice Questions (MCQ):

1 point
An inner product on R^3 R3 is defined as:

$\langle ., . \rangle \langle ., . \rangle$: $R^3 \times R^3$ R3 × R3 $\rightarrow R \rightarrow$ R

$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle$⟨(x1, x2, x3), (y1, y2, y3)⟩ = $x_1 y_1 + x_2 y_2 + x_3 y_3$ x1y1 + x2y2 + x3y3.

Match a set SS in Column A with the set TT given in column B, so that $T \subset Span(S)^{\perp}$ T ⊂ Span(S)⊥. Match a set in column B with the property it satisfies in column C.



Which of the following option are true?

- [ ] d → → ii → → 3
- [ ] d → → iii → → 4
- [ ] c → → ii → → 4
- [ ] a → → iv → → 1
- [ ] a → → ii → → 3
- [ ] b → → iv → → 2

No, the answer is incorrect.
Score: 0
Accepted Answers:
d → → iii → → 4
a → → ii → → 3

Suppose W_1 and W_2 W1 and W2 are subspaces of a vector space V V . Let P_{W_1} and P_{W_2} PW1 and PW2 denote the projection from V V to W_1 W1 and V V to W_2 W2 respectively and consider the following statements:

- **Statement P**: If $P_{W_1} + P_{W_2}$ PW1 + PW2 is a projection from V V to $W_1 + W_2$ W1 + W2, then $P_{W_1} \circ P_{W_2} + P_{W_2} \circ P_{W_1} = 0$.PW1 ∘ PW2 + PW2 ∘ PW1 = 0.

- **Statement Q**: $P^2_{W_1} := P_{W_1} \circ P_{W_1}$ PW12 := PW1 ∘ PW1 is not a projection from V V to W_1 W1.

- **Statement R:** If AA is the matrix representation of P_{W_2} PW2, then AA can not be a symmetric matrix.
- **Statement S**: $P_{W_1} - P_{W_2}$ PW1 − PW2 is a projection from VV to $W_1 + W_2$W1 + W2.

Find the number of correct statements

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
Which of the followings option(s) is (are) true?

☐ Let $u, v,$ and wu, v, and w be three vectors in R^2 R2 and $\langle .,. \rangle \langle .,. \rangle$ be an inner product on R^2 R2. If $\langle u, v \rangle = \langle u, w \rangle \langle u, v \rangle = \langle u, w \rangle$, then $v = w$v = w

☐ Let VV be an inner product space. If $\langle u, v \rangle = 0 \langle u, v \rangle = 0$ for all $u \in V$u ∈ V , then $v = 0$v = 0

☐ Let VV be an inner product space and $u, v \in V$u, v ∈ V . If $\langle u + v, u - v \rangle = 0 \langle u + v, u - v \rangle = 0$, then $\|u\| \neq \|v\| \|u\| = \|v\|$

☐ Let VV be an inner product space. If $S \subset V$S ⊂ V then $S^{\perp} = (span(S))^{\perp}$ S⊥ = (span(S))⊥

☐ Suppose $\{u_1, u_2, \ldots u_m\}$ {u1, u2, … um} be an orthogonal set of vectors in an inner product space V V , then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2 \|u1 + u2 + \ldots + um\|2 = \|u1\|2 + \|u2\|2 + \ldots + \|um\|2$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let VV be an inner product space. If $\langle u, v \rangle = 0 \langle u, v \rangle = 0$ for all $u \in V$u ∈ V , then $v = 0$v = 0
Let VV be an inner product space. If $S \subset V$S ⊂ V then $S^{\perp} = (span(S))^{\perp}$ S⊥ = (span(S))⊥
Suppose $\{u_1, u_2, \ldots u_m\}$ {u1, u2, … um} be an orthogonal set of vectors in an inner product space V V , then $\|u_1 + u_2 + \ldots + u_m\|^2 = \|u_1\|^2 + \|u_2\|^2 + \ldots + \|u_m\|^2 \|u1 + u2 + \ldots + um\|2 = \|u1\|2 + \|u2\|2 + \ldots + \|um\|2$.
1 point
Which of the following option(s) is/are true?

☐ Let AA and BB be orthogonal matrices of order 3. Then A and BA and B are equivalent matrices.

☐ Let AA and BB be orthogonal matrices. Then A and BA and B are similar matrices.

☐ Let AA be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θθ. Then nullity of the matrix AA is 0.

☐ Let AA be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θθ and BB be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X -axis with angle ββ. Then AA and BB are equivalent matrices.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let AA and BB be orthogonal matrices of order 3. Then A and BA and B are equivalent matrices.
Let AA be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θθ. Then nullity of the matrix AA is 0.
Let AA be a rotation matrix corresponding to the anti-clock wise rotation of the XY-plane about the Z-axis with angle θθ and BB be a rotation matrix corresponding to the anti- clock wise rotation of the YZ-plane about the X -axis with angle ββ. Then AA and BB are equivalent matrices.

Numerical Answer Type:

Let u = (1, 2, 1)u = (1, 2, 1) and v = (1, 2, 3)v = (1, 2, 3) be the vectors of the inner product space R^3 R3 with usual inner product. Let $T: R^3 \rightarrow R^3$ T : R3 → R3 be an orthogonal linear transformation. If θθ is an angle between $T(u)$T (u) and $T(v)$T (v), then find the value of $||T(u)||||T(v)||\cos\theta$ ||T (u)||||T (v)|| cos θ.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 8
1 point
Consider a set W = {(1, 1, 1)}W = {(1, 1, 1)} from the inner product space R^3 R3. Find the dimension of the subspace $W^\perp$ W ⊥, where $W^\perp$ W ⊥ is the collection of the vectors which are orthogonal to the vector (1,1,1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
Let v = (3, 1, 2)v = (3, 1, 2) be a vector in R^3 R3. If (a, b, c)(a, b, c) is the vector obtained from vv after the anti- clock wise rotation of ZX-plane with angle 45° 45∘ about the Y- axis, then find the value of $\sqrt{2}(a + b + c - 1)$ $\sqrt{2}$(a + b + c − 1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point

Comprehension Type Question:

With a particular frame of reference (in R^2 R2) the position of Rahul's house is at the point (8,1). There are two roads Road 1 and Road 2 along the lines $y = 2x$ and $y = -2x$y = 2x and y = −2x respectively on the XY − XY −plane.

Answer questions 8,9 and 10 using the given information.

1 point

Which of the following option(s) is (are) true?

- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).
- [ ] If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at $\frac{6}{5}(1, -2)56(1, -2)$.
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)56(1, -2)$.
- [ ] If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at (2,4).

No, the answer is incorrect.

Score: 0

Accepted Answers:

If Rahul wants to travel the minimum distance from his house to Road 1, then he will meet Road 1 at (2,4).

If Rahul wants to travel the minimum distance from his house to Road 2, then he will meet Road 2 at $\frac{6}{5}(1, -2)56(1, -2)$.

Let $\theta\theta$ be the acute angle between Road 1 and Road 2, then what will be the value of $5\cos(\theta)5\cos(\theta)$?

No, the answer is incorrect.

Score: 0